TRAFFIC LIGHT CONTROLLER

Verilog HDL Source Code — Timer-Based and Sensor-Based Designs

1. Timer-Based Traffic Light Controller

Source module: **Traffic_Light_Time**. Based on Appendix A of the supplied project report.

```
`timescale 1ns / 1ps

module Traffic_Light_Time(
    input wire clk,
    input wire clr,
    output reg [5:0] lights
);

    reg [2:0] state;
    reg [3:0] count;

    parameter S0 = 3'b000, S1 = 3'b001, S2 = 3'b010,
              S3 = 3'b011, S4 = 3'b100, S5 = 3'b101;

    parameter SEC5 = 4'b1111, SEC1 = 4'b0011;

    // State and counter update
    always @(posedge clk or posedge clr)
    begin
        if (clr == 1)
        begin
            state <= S0;
            count <= 0;
        end
        else
            case (state)
                S0: if (count < SEC5) begin state <= S0; count <= count + 1; end
                    else              begin state <= S1; count <= 0;         end
                S1: if (count < SEC1) begin state <= S1; count <= count + 1; end
                    else              begin state <= S2; count <= 0;         end
                S2: if (count < SEC1) begin state <= S2; count <= count + 1; end
                    else              begin state <= S3; count <= 0;         end
                S3: if (count < SEC5) begin state <= S3; count <= count + 1; end
                    else              begin state <= S4; count <= 0;         end
                S4: if (count < SEC1) begin state <= S4; count <= count + 1; end
                    else              begin state <= S5; count <= 0;         end
                S5: if (count < SEC1) begin state <= S5; count <= count + 1; end
                    else              begin state <= S0; count <= 0;         end
                default: state <= S0;
            endcase
    end

    // Output decoding
    always @(*)
    begin
        case (state)
            S0: lights = 6'b100001; // red green
            S1: lights = 6'b100010; // red yellow
            S2: lights = 6'b100100; // red red
            S3: lights = 6'b001100; // green red
            S4: lights = 6'b010100; // yellow red
            S5: lights = 6'b100100; // red red
            default: lights = 6'b100001; // red green
        endcase
    end

endmodule
```

TRAFFIC LIGHT CONTROLLER

Verilog HDL Source Code

2. Sensor-Based Traffic Light Controller

Source module: **Traffic_Light_Sensor**. Based on Appendix B of the supplied project report.

```
`timescale 1ns / 1ps

module Traffic_Light_Sensor(
    input clk,
    input clr,
    input SNS1,
    input SNS2,
    output reg [1:0] TL1, TL2
);

    reg [1:0] state, nextstate;

    parameter S0 = 2'b00, S1 = 2'b01, S2 = 2'b10, S3 = 2'b11;
    parameter green = 2'b00, yellow = 2'b01, red = 2'b10;

    // State register
    always @(posedge clk, posedge clr)
    begin
        if (clr) state <= S0;
        else     state <= nextstate;
    end

    // Next-state logic
    always @(*)
    begin
        case (state)
            S0: if (SNS2) nextstate = S1;
                else      nextstate = S0;
            S1: nextstate = S2;
            S2: if (SNS1) nextstate = S3;
                else      nextstate = S2;
            S3: nextstate = S0;
            default: nextstate = S0;
        endcase
    end

    // Output logic
    always @(*)
    begin
        if (state == S0)
        begin
            TL1 = green;  TL2 = red;
        end
        else if (state == S1)
        begin
            TL1 = yellow; TL2 = red;
        end
        else if (state == S2)
        begin
            TL1 = red;    TL2 = green;
        end
        else
        begin
            TL1 = red;    TL2 = yellow;
        end
    end

endmodule
```

Note: The supplied report shows testbench stimulus as separate excerpts. This PDF contains the two complete controller modules, not a standalone complete testbench.